NUS-ISS

Problem Solving Using Pattern Recognition

Deep learning: Identify

by Zhengqing Hu

© 2024 National University of Singapore.
All Rights Reserved.

© 2024 National University of Singapore. All Rights Reserved.

How deep learning is used to identify:
Use cases

© 2024 National University of Singapore. All Rights Reserved.

Amazon Go Stores: “Just Walk Out” Technology

State-of-the-art shopping experience

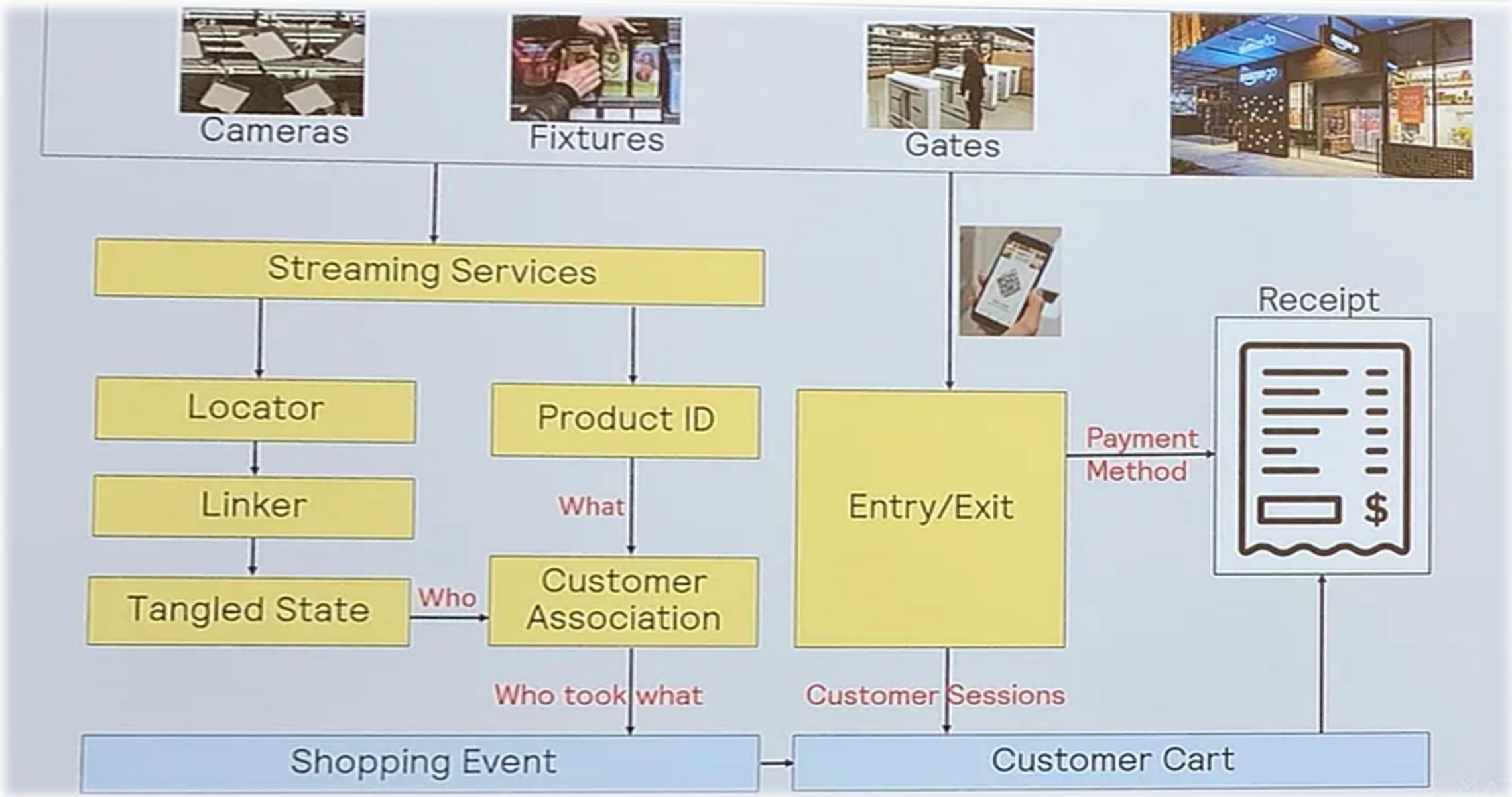


Source: https://www.youtube.com/watch?v=8hxJ8GbZ0Ok

© 2024 National University of Singapore. All Rights Reserved.

Amazon Go Stores: “Just Walk Out” Technology

Palm, Person & Product Identification



“Just Walk Out” Architecture

Source: https://towardsdatascience.com/how-the-amazon-go-store-works-a-deep-dive-3fde9d9939e9

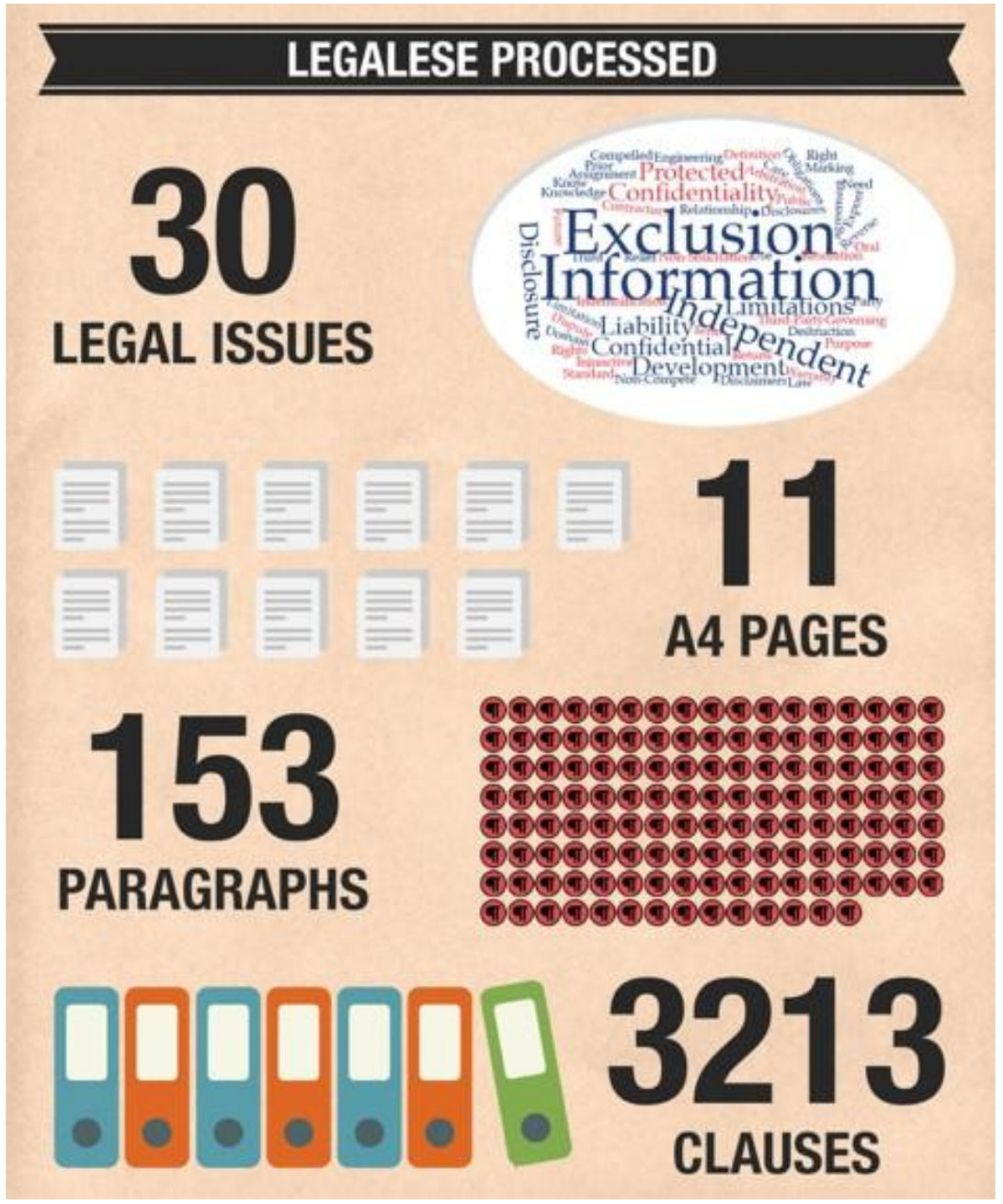
© 2024 National University of Singapore. All Rights Reserved.

Lawyers vs AI

Spotting the risk in NDA



20 lawyers with experience reviewing contracts at compaines including Goldman Sachs, Cisco and etc.

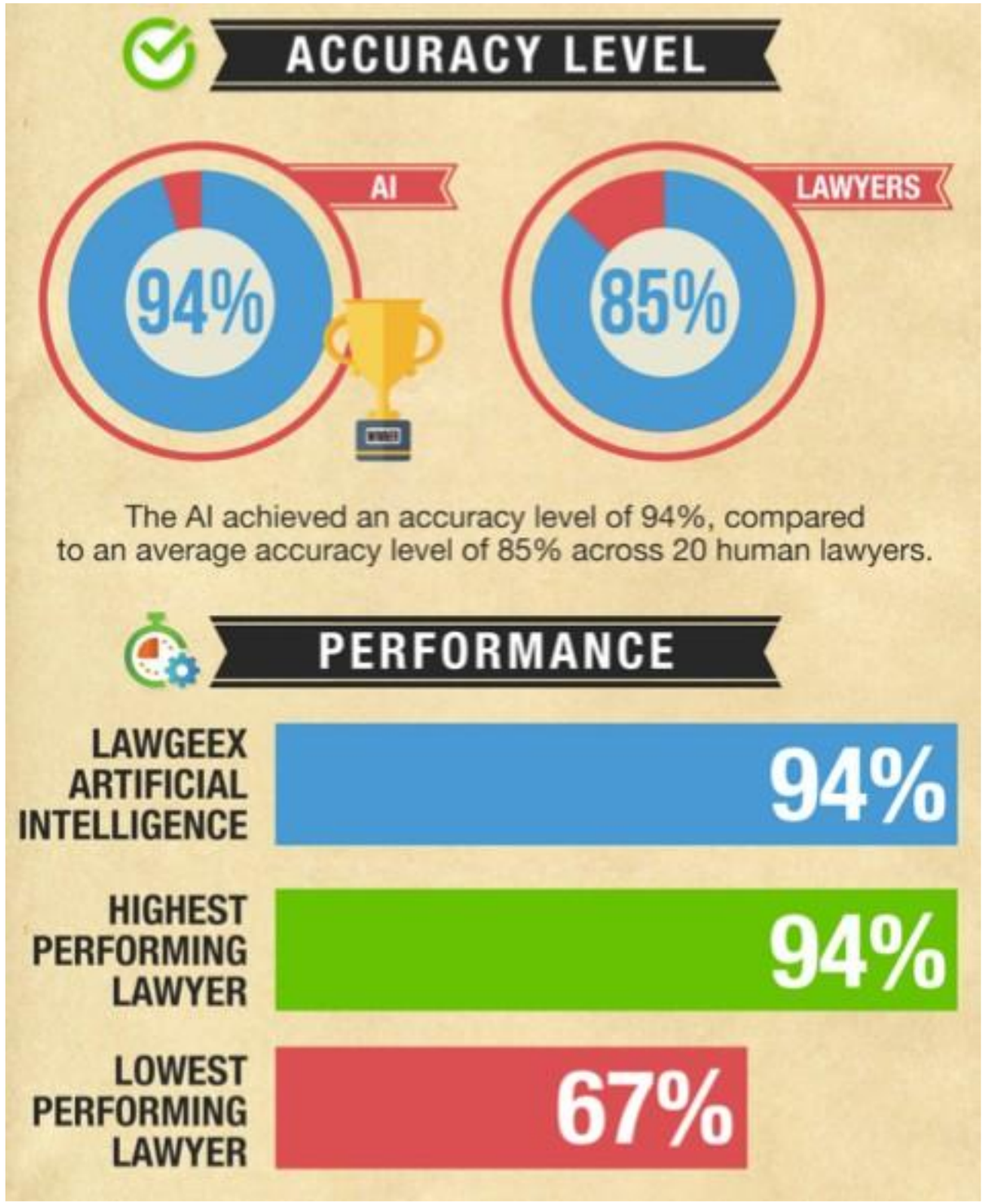


Source:
https://www.artificiallawyer.com/2018/02/26/lawgeex-hits-94-accuracy-in-nda-review-vs-85-for-human-lawyers/

© 2024 National University of Singapore. All Rights Reserved.

Lawyers vs AI

Spotting the risk in NDA



It took an average of **92 minutes** for the lawyers to review all 5 NDAs. In contrast, it took the AI a total time of **26 seconds** to review all 5 NDAs

Source:
https://www.artificiallawyer.com/2018/02/26/lawgeex-hits-94-accuracy-in-nda-review-vs-85-for-human-lawyers/

© 2024 National University of Singapore. All Rights Reserved.

To spot unfair clauses?

Claudette comes to rescue

- Visit http://claudette.eui.eu/tools/index.html

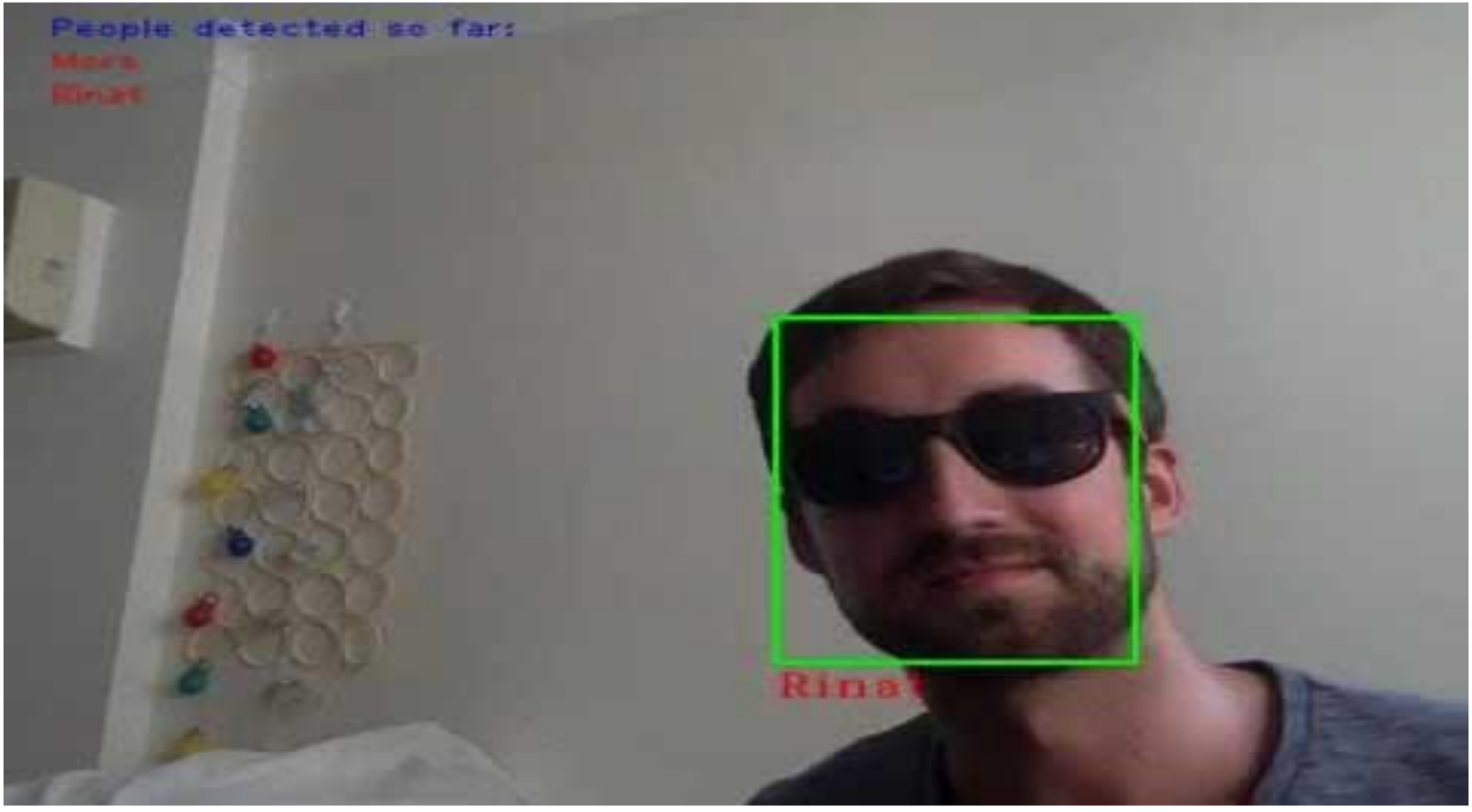


Source: https://arxiv.org/pdf/1805.01217.pdf

© 2024 National University of Singapore. All Rights Reserved.

FaceNet: Facial Recognition

By Google



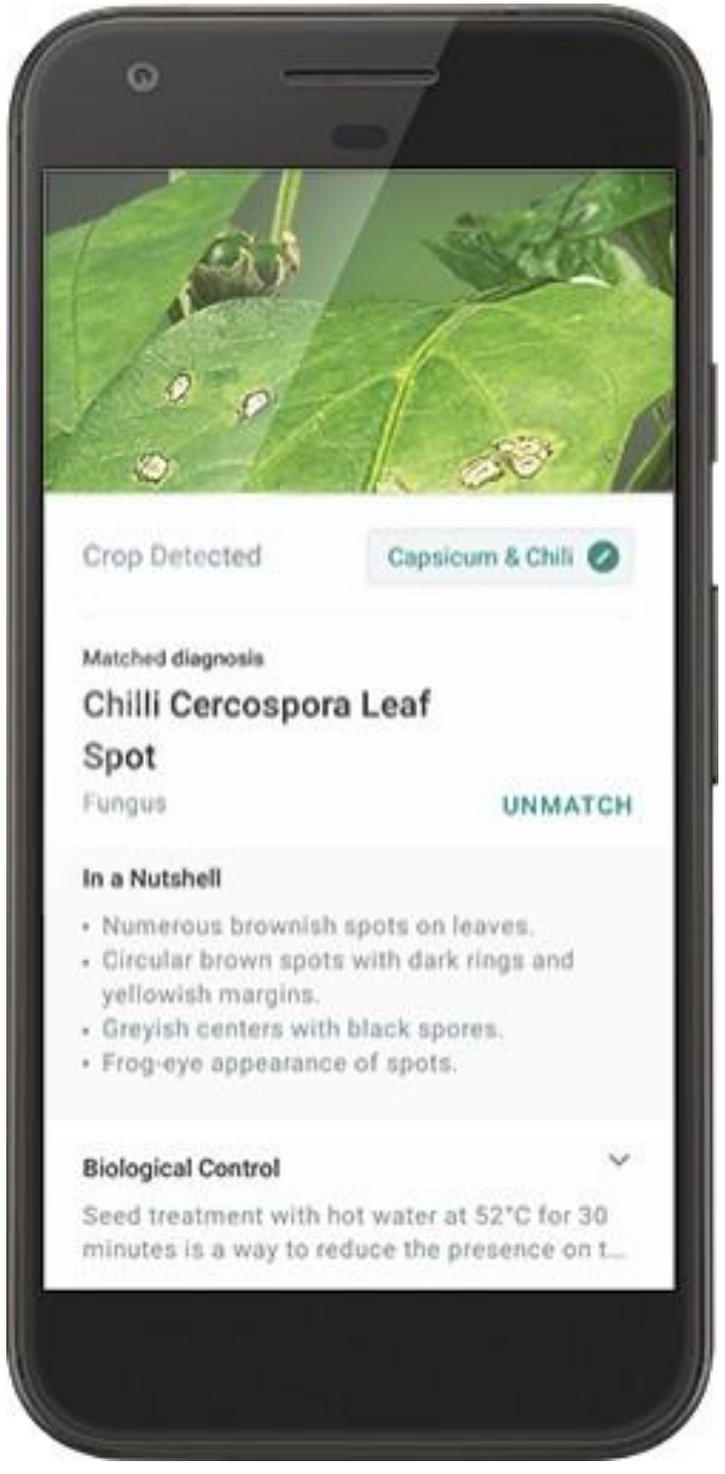
Source: https://arxiv.org/pdf/1503.03832

© 2024 National University of Singapore. All Rights Reserved.

Plants' disease

Better detection





Source: https://plantix.net/en

© 2024 National University of Singapore. All Rights Reserved.

DeepX: Security and Surveillance

Suspicious Behaviour, PPE Compliance and Trespassing Identification,
Facial Recognition for Access Control



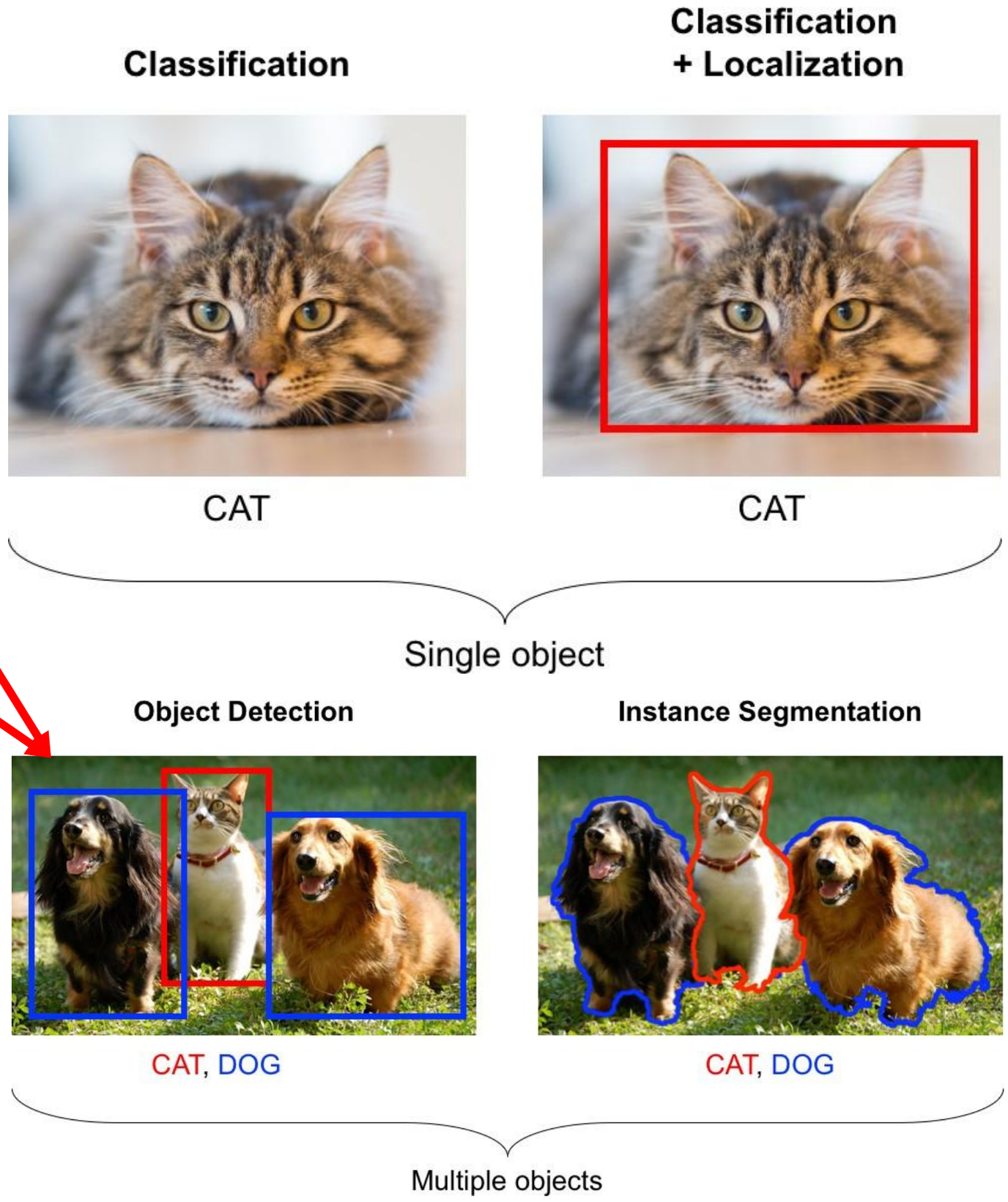
Source: https://deepxhub.com/solutions/;
https://www.youtube.com/watch?v=HHXRqCGCRCs

© 2024 National University of Singapore. All Rights Reserved.

When it comes to vision ...

A few things to note

- **Classification by itself will get confused if there are other objects inside the image**
- **Hence, we require localization to draw a bounding box (or bbox) to separate the target object and other objects**



Source: Deep Learning for Computer Vision by Rajalingappaa Shanmugamani

© 2024 National University of Singapore. All Rights Reserved.

Segment Anything: Identify Segmentations

Research by Meta AI

Utilizes an advanced deep learning technique known as Vision Transformers (ViT), an alternative to traditional Convolutional Neural Networks



Bounding box prompts from an object detector can enable text-to-object segmentation.



SAM can take input prompts from other systems, such as in the future taking a user's gaze from an AR/VR headset to select an object

Source: https://segment-anything.com/;
https://arxiv.org/pdf/2304.02643

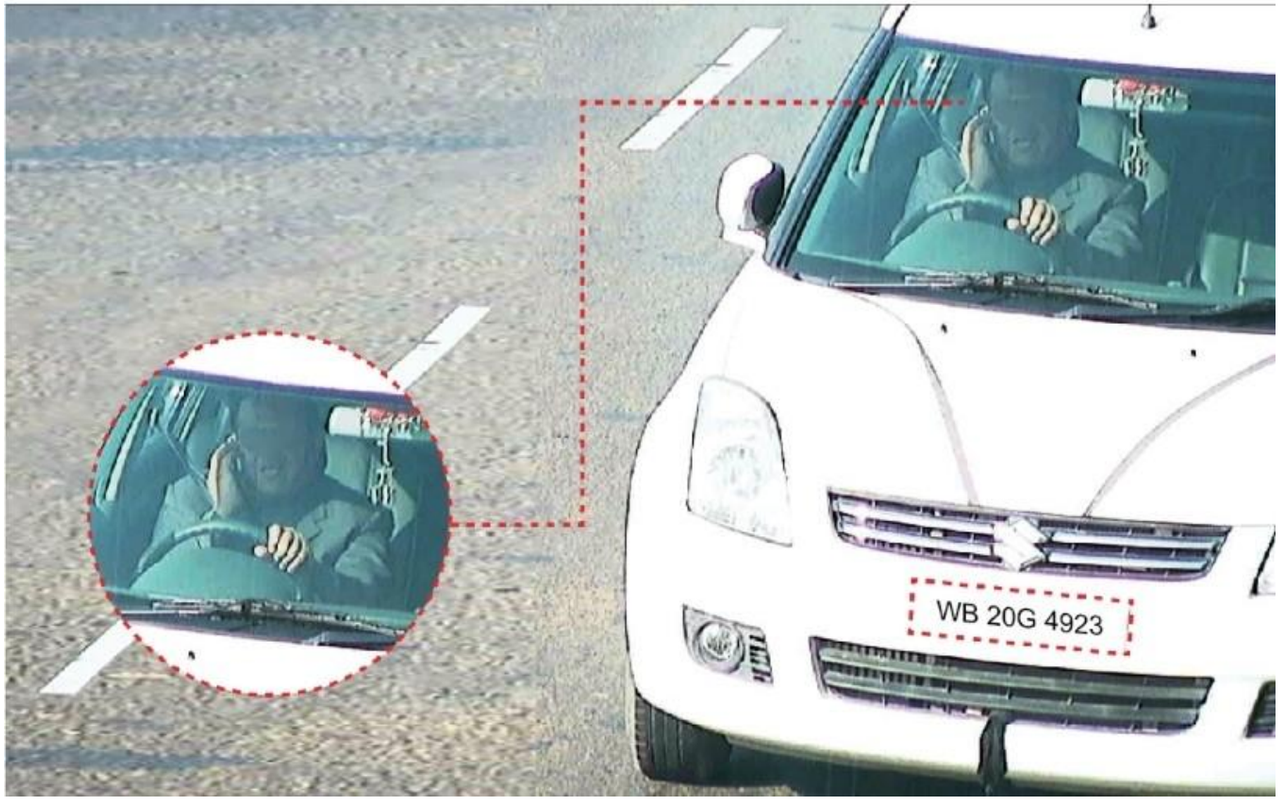
© 2024 National University of Singapore. All Rights Reserved.

Object detection

Identify traffic offences

- NO seat belt detection
- Using-phone-while-driving detection





Source: https://www.sourcesecurity.com/news/videonetics-artificial-intelligence-deep-learning-no-seat-belt-detection-co-4403-ga-npr.1551018780.html

© 2024 National University of Singapore. All Rights Reserved.

Object detection

For autonomous vehicle



Source: https://www.youtube.com/watch?v=chzq2E75M84

© 2024 National University of Singapore. All Rights Reserved.

Volvo in-car cameras

Combat distraction and drunk driving

- 'Pro-active' driver monitoring system



Source: https://www.caranddriver.com/news/a26893035/volvo-interior-cameras-distraction-drunk-driving/

© 2024 National University of Singapore. All Rights Reserved.

Aircraft

Monitoring check-in

Learn more



Source: https://www.assaia.com/solutions/apron-ai

© 2024 National University of Singapore. All Rights Reserved.

Behaviour identification

by Aipoly



Source: https://www.aipoly.com;
https://www.youtube.com/watch?v=t33G77R7FJQ

© 2024 National University of Singapore. All Rights Reserved.

Object detection

Auto referee?

- Track ball and players
- Report score based on the events in scene



Source: https://arxiv.org/abs/2004.09927;
https://www.youtube.com/watch?v=vLgZBOOVzP4

© 2024 National University of Singapore. All Rights Reserved.

Quick Exercise: “Identify” Use Case using Deep Learning

© 2024 National University of Singapore. All Rights Reserved.

Quick Exercise (10 mins)

The previous use cases are examples that illustrate the utilization of deep learning for identification purposes.

<u>You are to:</u>

- **List out 1 unique and interesting use case that utilizes deep learning for identification purposes**
- The only condition is that your input must not be mentioned before in this module so far
- Feel free to elaborate more on the concept of your use case via pictures or videos, etc

Put your research findings in PowerPoint Slides and upload the slide file (PPT or PDF allowed). Lucky individuals will be chosen randomly to present their findings.

© 2024 National University of Singapore. All Rights Reserved.